

# *Modern Real- Time* LAN Chat SYSTEM

REAL-TIME COMMUNICATION & USER TRACKING

Presented by  
**Vivek Kumar Gupta**

Project  
**User Tracking  
System**



# *Introduction* to LAN Chat

## **What is LAN Chat?**

*Real-Time Local Messaging*

**Traditional communication tools often rely on internet connectivity, leading to delays and security concerns. LAN chat eliminates these challenges.**

**A LAN Chat System enables real-time messaging within local networks, allowing devices to communicate instantly without internet dependency. This is crucial for modern workplaces, educational institutions, and secure environments where fast, private communication is essential.**





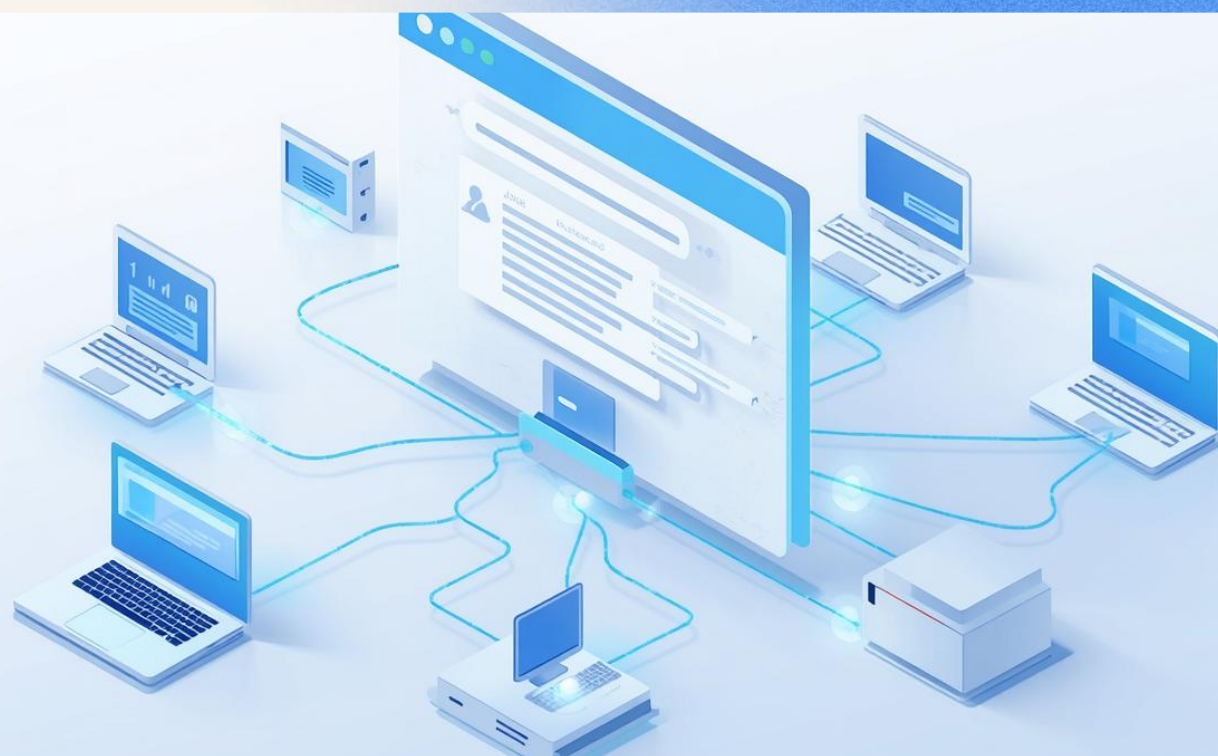
LAN Chat System

# Project Overview

Our system is a browser-based LAN communication platform that enables real-time messaging powered by WebSocket technology through Socket.IO. The event-driven architecture ensures instant updates and seamless communication across all connected devices.

Designed for secure, internet-independent operation, the platform delivers fast and reliable messaging entirely within your local network infrastructure.

The system prioritizes privacy and performance by keeping all communication local. With no external internet dependency, users enjoy minimal latency, enhanced security, and complete control over their messaging environment.



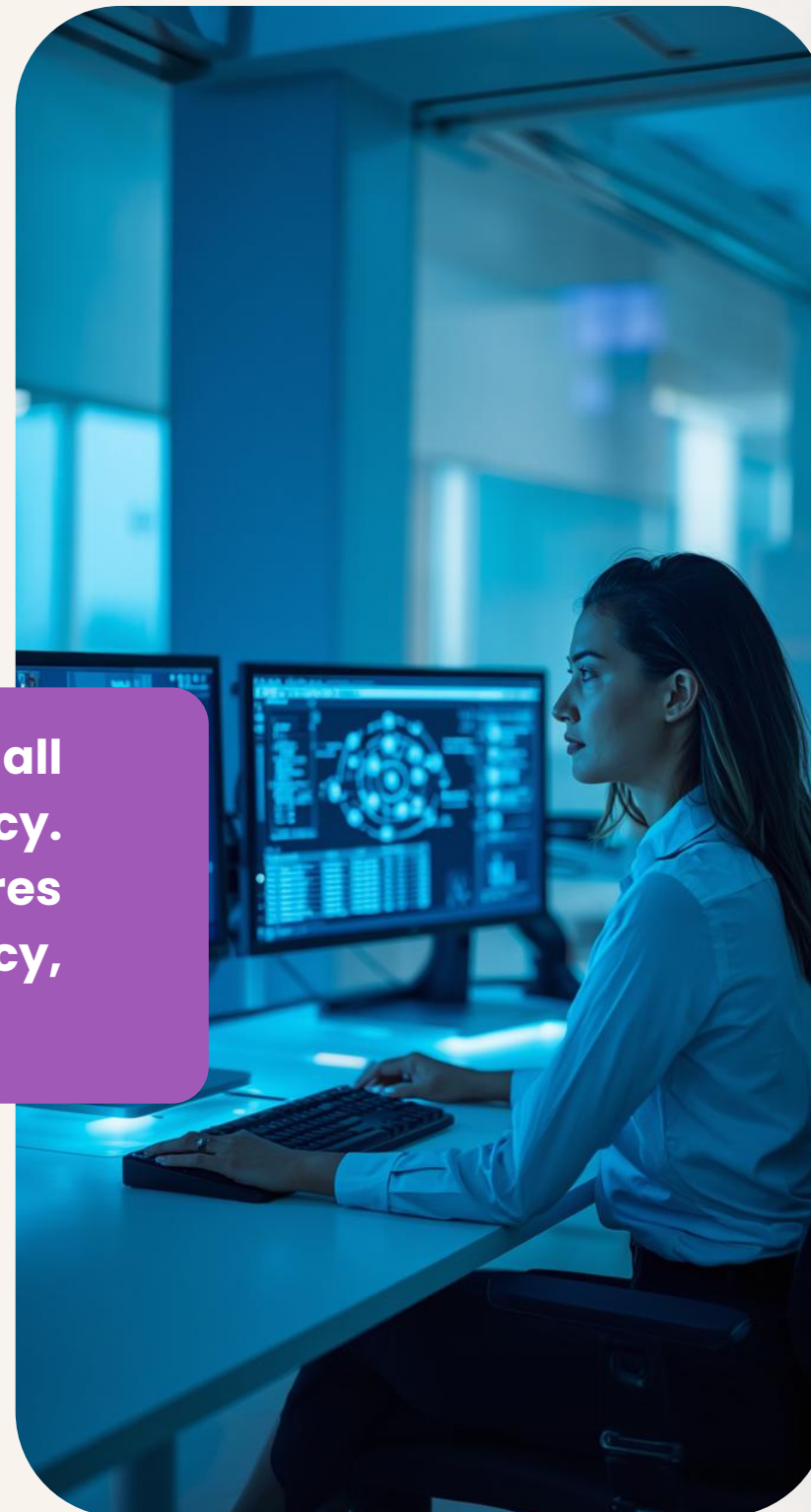




# Objectives

**Enable lightning-fast messaging across all LAN-connected devices with minimal latency. Our WebSocket-powered architecture ensures instant delivery without internet dependency, keeping your team connected seamlessly.**

**Monitor active users and presence indicators in real-time. Track who's online, typing, or away with live status updates across your local network.**



**Fast LAN  
Communication**



**Real-Time User  
Tracking**



**Instant Updates Offline**



**Secure Local Privacy**





# Technologies Used



## Frontend Stack

HTML5 for structure, CSS3 for styling with glassmorphism effects, JavaScript for dynamic interactions and real-time functionality.

## Backend Stack

Node.js runtime environment, Express.js web framework, Socket.IO for WebSocket-based real-time communication.

## Database Layer

MongoDB NoSQL database for flexible data storage, Mongoose ODM for elegant schema modeling and validation.

## Development Tools

VS Code as primary IDE for development, MongoDB Compass for visual database management and queries.



# System Architecture



## Client-Server Model

Devices connect directly to a central server over LAN using WebSocket protocol via Socket.IO for instant bidirectional communication.

100%

WebSocket Events

<1ms

Data Exchange

24/7

Persistence



MongoDB provides persistent storage for message history and user data, ensuring chat continuity across sessions.



# Key Features

Modern glassmorphism feature cards with glowing neon icons showcase instant messaging, live presence detection, real-time typing status, and seamless voice note and file sharing capabilities across your local network.



Real-time  
messaging



User tracking  
& presence



Typing indicator



Voice notes &  
file sharing





# Real-Time Communication Flow



## Event Lifecycle



Socket.IO manages connection, message emission, and broadcasting events seamlessly for instant real-time communication across all connected LAN devices.



## Typing Indicators

Real-time typing events notify all participants instantly when someone is composing a message, creating a responsive and engaging chat experience.







# *Database Integration*

**Our system leverages MongoDB and Mongoose for robust message storage architecture. Every conversation is persistently stored, enabling seamless chat history retrieval even when users reconnect after being offline.**

**User data and device tracking information are securely stored in the database, ensuring reliable presence detection and session management across the local network.**

**The event-driven architecture ensures real-time synchronization between MongoDB and connected clients, with data flow arrows seamlessly connecting server operations to user interfaces for instant updates.**



# UI/UX Design



**Glassmorphism UI with frosted glass effects creates a modern aesthetic. WhatsApp and Telegram inspired chat interface ensures familiarity. Dark theme with blue and purple neon accents provides visual appeal.**

**User avatars, live timestamps, and emoji support enhance engagement. Animated backgrounds with subtle particle effects create an immersive experience.**





# Screenshots of the System



## Chat Screen Interface

Real-time message bubbles with typing indicators, live timestamps, and emoji support. Dark theme with glassmorphism effects and neon blue accents for modern appeal.

## User List & Presence

Dynamic user list showing real-time online/offline status with animated presence indicators and user avatars.

## QR Code & File Sharing

Seamless QR code connection interface for quick device pairing. Integrated file sharing and voice note UI with media preview modals and progress indicators.



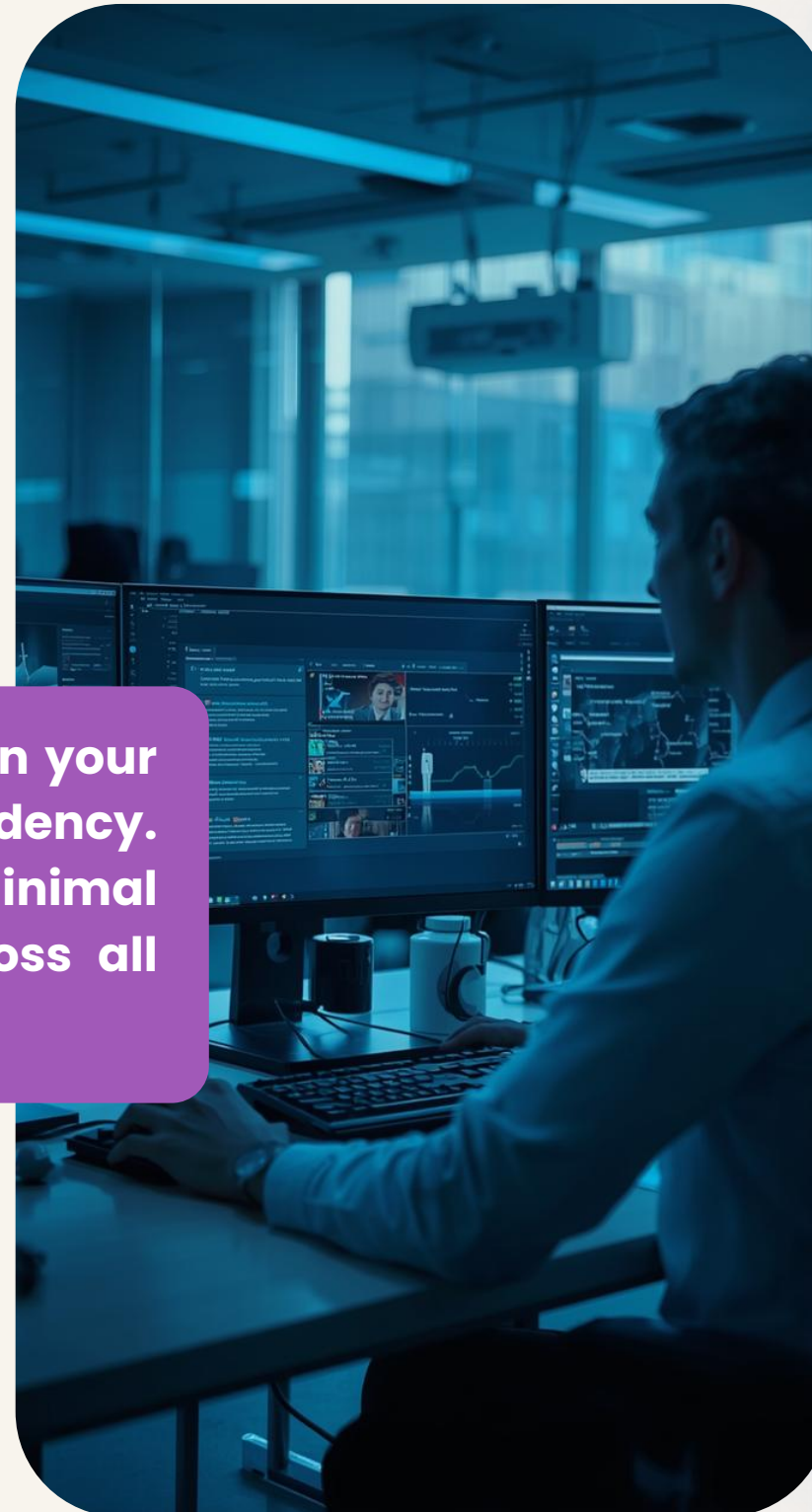


LAN Chat System

# Advantages

**Our LAN chat system operates entirely within your local network, eliminating internet dependency. Enjoy lightning-fast communication with minimal latency and instant message delivery across all connected devices.**

**Easy deployment and maintenance with secure, private communication that stays within your network boundaries.**



**No Internet  
Required**



**Fast &  
Lightweight**



**Real-Time  
Updates**



**Privacy Focused**





# Scope

## Video Calling

Integrate real-time video calling capabilities for face-to-face communication within the LAN environment, enabling richer collaboration.

## E2E Encryption

Implement end-to-end encryption protocols to ensure all messages and files remain completely private and secure.

## AI Chatbot Integration

Add intelligent AI chatbot features for automated responses, smart suggestions, and enhanced user assistance.

## Cloud Sync & Mobile App

Enable cloud synchronization and develop a dedicated mobile application for seamless cross-platform access.





# Conclusion



**Our Modern Real-Time LAN Chat System delivers fast, secure, and instant communication across local networks. With WebSocket-powered messaging and event-driven architecture, users experience zero-latency updates without internet dependency.**

**Privacy remains paramount with fully local LAN operation, ensuring sensitive communications never leave your network. The system operates independently, providing reliable connectivity even during internet outages.**

**Built with a modern glassmorphism UI and scalable MongoDB architecture, the platform combines aesthetic excellence with technical robustness. From real-time presence tracking to file sharing, this system represents the future of local network communication.**





Thank You!

QUESTIONS?

*We welcome your questions and feedback. Connect with us for more information about the Modern Real-Time LAN Chat and User Tracking System.*

## Submitted By

**Name:** Vivek Gupta

**College:** Arka Jain University,  
Jamshedpur

**Department:** Computer Science and  
Engineering

**Academic Session:** 2025-2026

**Project Type:** Networking Based  
Web Application



8210457599



vivekgupta821045@gmail.com



Jamshedpur, Jharkhand